Justin **McMillen**

jomcmillen@gmail.com | xxx | Tampa, FL (open to relocation) | Google Scholar | LinkedIn GitHub

Education

University of South Florida — *Tampa, FL, USA*
Ph.D. of Electrical Engineering (Advisor: Dr. Yasin Yilmaz) — *Aug. 2022 - Present (Expected May 2027)*

Research Focus: World Models, Object-Centric Representation Learning, Self-Supervised Learning

University of South Florida — *Tampa, FL, USA*
M.S. of Electrical Engineering — *Aug. 2022 - May 2025*

University of South Florida — *Tampa, FL, USA*
B.S. of Electrical Engineering — *Aug. 2018 - May 2022*

Fukuoka Seikei Gakuen — *Fukuoka, Japan*
A.A. of Japanese Communication — *Apr. 2016 - May 2018*
Japanese Language Proficiency Test N1 (Fluent)

Research Experience

Research agenda: the core components an autonomous agent needs to act over long horizons in a changing, partially observed world: **perception** (SP-JEPA), **memory** (COMPASS), and **metacognition** (PONDER), each studied in isolation with the other components as oracles.

Memory: Object Persistence in World Models (COMPASS) — *2026 - Present*

- Reproduced FloWM (ICML 2026), a latent world model with a velocity-shifted grid memory, to its published prediction error (MSE 0.000603); rebuilt its unreleased position probe and data generator, fixing four generator bugs.
- In a pre-registered, independently replicated test on 1,024 held-out videos with off-basis motion, an object-node memory (Kalman tracker on oracle detections) kept 94% of out-of-view objects vs. 4% for FloWM ($\leq$13% after fine-tuning), in 14 KB of state vs. 5.6 MB.
- Showed that memory scores depend on the reader and on rendering: a single-cell probe finds 66% of out-of-view objects where FloWM's probe finds 89%, and attributed FloWM's 7.5-point in-view lead to writing its own predicted frames back into memory.
- Building a learned object-node memory (per-object position and velocity beliefs, re-identification by appearance) that scales with object count, not room area (FloWM: 15.5 GiB at 16 cells, a projected 169 GiB at 60).

Perception: A Controlled Study of Superpixel Tokens in JEPAs (SP-JEPA) — *2026 - Present*

- Built SP-JEPA, an I-JEPA-style ViT that reads SLIC superpixel tokens and trains with cross-view latent prediction, alongside a grid twin that differs only in the partition; trained an 18-run controlled matrix spanning ViT-Tiny to ViT-Base and ImageNet-100 to ImageNet-1K.
- Designed an encoder-by-readout protocol that separates what an encoder learns from the map it is read out on; on VOC (3 training $\times$ 5 probe seeds), neither encoder led by the pre-registered bound, and grid features read out on superpixels matched superpixel features.
- Identified superpixel geometry as a shortcut under latent prediction: the predictor consumes target shape as a partial view of the target (swapping it raises loss 21–24% with no better fit), and writing shape into tokens costs 3.4–5.0 ImageNet-100 points.

Metacognition: When Should an Agent Ask an LLM Advisor? (PONDER) — *2026 - Present*

- Built a Craftax agent stack (counterfactual-rule worlds, a planner, an exactly replayable LLM advisor) to study when an agent should pay to ask a frozen LLM; in a pre-registered recheck, asking raised diamond success by 11 points (200 paired episodes, $p = 0.004$) but hurt where the advice was wrong.
- A power analysis on $\sim$179K real decision states showed, before committing cluster compute, that a learned question policy could not separate an answer's value from noise; redirected into a controlled study.

University of Tokyo — *Tokyo, Japan*

Visiting Researcher, Machine Intelligence Lab (Prof. Tatsuya Harada) — *July 2026*

- Gave a formal seminar on SP-JEPA to Prof. Harada's group, followed by a technical discussion of tokenizer design and positional embeddings for object-centric world models.

Earlier Research: Multimodal and 3D Perception *2022 - 2025*

- **Permafrost thaw mapping from lidar:** projection decoder with learned height embeddings and z-stratified sampling that preserves vertical forest structure when projecting drone-lidar point clouds to 2D, on a new dataset from interior Alaska.
- **FuseForm:** local-global fusion transformer for multimodal semantic segmentation; outperformed state-of-the-art models across 5 datasets by 2–15% (WACV 2025 GeoCV Workshop).
- **SegGen:** Unreal Engine 5 pipeline with an autonomous drone-collection system for pixel-accurate multimodal segmentation datasets; released as an open-source tool and dataset (MDPI Sensors 2025).
- **RF fingerprint authentication:** deep learning pipeline on I/Q samples from chaotic antenna arrays, reaching 100% device classification across up to 300 users (WAMICON 2023, invited; IEEE Access 2025).

Publications

Grid or Nodes? Probing Object Memory in World Models Under Partial Observability *2026*
Under Review, CoRL 2026 Workshop: Do Robots Need World Models?
J. McMillen & Y. Yilmaz

Do Superpixel Tokens Change What a JEPA Learns? A Controlled Study *2026*
In Preparation for CVPR 2027
J. McMillen & Y. Yilmaz

Preserving Vertical Structure in 3D-to-2D Projection for Permafrost Thaw Mapping *2026*
In Preparation for IEEE Transactions on Geoscience and Remote Sensing (TGRS)
J. McMillen, R. Van Alphen, T. Sadeghi Chorsi, J. Shabaga, M. Rodgers, R. Malservisi, T. Dixon & Y. Yilmaz

FuseForm: Multimodal Transformer for Semantic Segmentation *2025*
GeoCV Workshop at Winter Conference on Applications of Computer Vision (WACV) 2025
J. McMillen & Y. Yilmaz

SegGen: An Unreal Engine 5 Pipeline for Generating Multimodal Semantic Segmentation Datasets *2025*
MDPI Sensors 2025, 25(17), 5569
J. McMillen & Y. Yilmaz

Hardware and Deep Learning Based Authentication through Enhanced RF Fingerprinting *2025*
IEEE Access 2025
J. McMillen, F. Razak, G. Mumcu, & Y. Yilmaz

Deep Learning-based RF Fingerprint Authentication with Chaotic Antenna Arrays *2023*
IEEE Wireless and Microwave Technology Conference (WAMICON) 2023, Invited Paper
J. McMillen & Y. Yilmaz

Professional Experience

SAS Institute - Advanced Analytics R&D *Cary, NC (Remote)*
IoT Machine Learning PhD Research Intern *May 2025 – Present*

- Investigating JEPA-based self-supervised learning for predictive maintenance of rotating machinery: spectrogram pretraining on vibration signals with a lightweight fault-classification head, plus per-deployment LoRA adaptation to reduce cold-start time.
- Designed and ran a large-scale evaluation of 56 hyperspectral image-processing algorithms (segmentation, feature extraction, dimensionality reduction) across 7 datasets; results inform product decisions.

Secure and Intelligent Systems Lab - University of South Florida *Tampa, FL*
Graduate Research Assistant, Lab Manager *Aug. 2022 – Present*

- Mentored 2 Ph.D. students and 1 undergraduate on funded projects (NSF, Army Corps of Engineers, DoD); manage the lab's 10 workstations and two GPU nodes (2×A100, 6×H100).

Tampa Electric *Tampa, FL*
Distribution Control Room Engineer (Intern) *Jan. 2020 – May 2022*

- Implemented genetic algorithm to solve systemwide phase imbalance, saving $2M and 1,000 tons of CO_2 emissions annually.

Awards & Honors

NSF Research Trainee Program (NRT) *2025 - Present*
University of South Florida - Electrical Engineering
Competitive 2-year, $68K fellowship from the National Science Foundation for high-priority convergent research.

Sigma Xi, Full Member *2025 - Present*
Scientific Research Honor Society

Skills

Programming	**Python** - PyTorch, NumPy, Pandas, OpenCV, scikit-learn **MATLAB**
Deep Learning	World Models, Self-Supervised Learning (JEPA, VICReg), Object-Centric Learning, Vision Transformers, Probing & Representation Analysis, State Estimation & Tracking (Kalman Filtering), Point Cloud Networks (PTv3), Multimodal Fusion
Methods	Pre-registered evaluation, controlled ablations, paired bootstrap CIs, power analysis
Tools	HPC/SLURM, Git/GitHub, Claude Code, Unreal Engine 5, LaTeX